System Response Mechanism

1. Introduction

The System Response Mechanism of the **AI-Based Mini SIEM** is designed to provide immediate, actionable intelligence upon the detection of security anomalies. Unlike passive logging systems, this module actively evaluates the severity of detected threats, suppresses redundant alerts to prevent fatigue, and generates comprehensive incident reports with tailored defensive countermeasures. The mechanism operates in real-time, integrating seamlessly with the detection engine and the dashboard visualization layer.

2. Threat Detection and Alert Generation

The core of the response system is the `AlertManager`, a dedicated logic engine that processes incoming log streams and anomaly scores. It functions as a decision-making layer between the raw data ingestion and the final user notification.

2.1 Severity Classification Strategy

To prioritize critical threats, the system employs a hybrid classification strategy combining machine learning anomaly scores with heuristic rule-based logic. Each detected event is assigned a severity level (`High`, `Medium`, `Low`, or `Normal`) and a specific threat type.

Threat Type	Severity	Detection Logic	Description
Brute Force Attempt	High	`login_status == 'FAILURE'`	Repeated failed login attempts indicating a credential stuffing or password cracking attack.
Data Exfiltration	High	`packet_size > 10000`	Abnormally large packet sizes suggesting unauthorized data transfer.
ICMP Tunneling	Medium	`protocol == 'ICMP' AND packet_size > 1000`	Use of ICMP packets (normally small) to carry large payloads, often used for covert channels.
Behavioral Anomaly	Medium	`anomaly_pred == 1`	Statistical deviation detected by the Ensemble Learning model (Isolation Forest, OCSVM, Autoencoder) without a specific rule match.

2.2 Intelligent Deduplication

To maintain operator efficiency, the system implements an intelligent deduplication mechanism. Repeated alerts from the same source IP address for the same threat type are suppressed within a dynamic window (default: 60 seconds). This ensures that a flood of similar packets—common in DoS attacks or brute force scripts—results in a single, consolidated alert rather than thousands of individual notifications.

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3. Automated Incident Reporting

Upon detecting a **High Severity** incident, the system automatically triggers the `ReportGenerator` module. This component creates a detailed, standardized incident report in Markdown format, stored in the `reports/` directory. This automation ensures that forensic evidence is preserved immediately, reducing response time.

Key Components of an Incident Report

- **Incident ID & Timestamp:** Unique identifier for tracking.
- **Threat Intelligence:** Source IP, Destination IP, and specific attack vectors.
- **Technical Analysis:** Detailed breakdown of the payload or behavior (e.g., protocol used, packet size).
- **Response Recommendations:** A set of tailored actions based on the specific threat type.

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4. Defensive Countermeasures

The system provides dynamic, context-aware recommendations to guide security analysts. These recommendations are generated based on the specific `Threat Type` identified by the classification engine.

Threat Scenario	Recommended Defensive Actions
Brute Force	1. **Block Source IP:** Immediate firewall block of the attacking IP. 2. **Reset Credentials:** Force password reset for affected accounts. 3. **Audit Logs:** Review successful logins from the IP to check for compromise.
Denial of Service (DoS)	1. **Rate Limiting:** Enable strict rate limiting on incoming traffic. 2. **Blackhole Traffic:** Route traffic from the source IP to a null interface. 3. **Scale Resources:** Temporarily increase server capacity to absorb the load.

Threat Scenario	Recommended Defensive Actions
Data Exfiltration	1. **Isolate Host:** Disconnect the affected host from the network immediately. 2. **DLP Scan:** Initiate Data Loss Prevention scans for sensitive data transfer logs. 3. **Forensic Analysis:** Capture memory and disk images for post-incident analysis.
Generic / Unknown	1. **Investigate:** Correlate with other logs and sources. 2. **Monitor:** Add the source IP to a high-priority watch list for enhanced logging.

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5. Passive vs Adaptive Response Strategy

The Mini SIEM system adopts a layered response strategy that combines passive monitoring with adaptive intelligence-driven responses.

5.1 Passive Monitoring Response

The system primarily operates as a monitoring and intelligence platform. Passive response mechanisms focus on detecting, analyzing, and reporting threats without directly blocking or modifying network traffic. These responses include:

- Generating security alerts based on anomaly detection and rule-based analysis.
- Visualizing detected threats in real-time dashboards.
- Producing automated incident reports for forensic investigation.
- Logging historical events for auditing and trend analysis.

This approach ensures that security analysts maintain full control over mitigation decisions while preserving detailed security evidence.

5.2 Adaptive Intelligence Response

In addition to passive monitoring, the system implements adaptive response capabilities that allow it to evolve with changing network behavior. These include:

- Detecting concept drift in network traffic patterns.
- Automatically retraining machine learning detection models.
- Improving detection accuracy over time through behavioral learning.
- This adaptive layer enables the system to remain effective against emerging and previously unseen threats.

5.3 Design Limitation and Scope

The Mini SIEM is designed as a detection and intelligence platform rather than an automated prevention or blocking system. Defensive actions such as firewall blocking, network isolation, or credential resets are provided as recommended countermeasures for human analysts rather than executed automatically. This design ensures operational safety and reduces the risk of unintended service disruption caused by automated blocking decisions.

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6. Visual Dashboard Integration

The response mechanism is tightly integrated with the system dashboard to provide real-time situational awareness.

- **Live Alert Feed:** A dedicated panel displays the most recent 50 alerts, color-coded by severity (Red for High, Orange for Medium).
- **Immediate Notifications:** High-severity alerts trigger instant "Toast" notifications on the dashboard, alerting the operator even if they are navigating other menus (e.g., Model Performance).
- **Drift Detection:** The system continuously monitors for "Concept Drift" in the data stream. If the statistical properties of the traffic change significantly (indicating a new normal or a stealthy attack pattern), the system alerts the user and suggests model retraining.